Separable Differential Equations with Initial Conditions Answer Key

AP Calculus AB/BC   Differential Equations

Quick Answers

1. $2x^3$, 17
2. $x^3 - 4x$, 5
3. $\frac{x^2}{2}$, 14.78
4. $\frac{9}{2}$, 4.58
5. $-\frac{1}{y}$, 2
6. 14.14
7. $\sin(x)$, 9.28
8. y^2, 5.48
9. 13.73
10. 0, 3

Curriculum

Level: AP Calculus AB/BC   Differential Equations
FUN-7 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*
Difficulty 1–5 of 5 across 18 tagged checks.

Common wrong answers

3. *If they got 7.39: dropped the initial condition and used a leading coefficient of 1 instead of 2*
4. *If they got 3.24: forgot to double both sides after integrating, using y^2 = x^2 + x + 4.5*
6. *If they got 20: rounded 25 days down to two whole half-lives and halved twice*
7. *If they got 1.72: integrated cosine to negative sine, giving a decaying exponential instead*
9. *If they got 91.42: took the logarithm of the target population instead of the ratio*

1. $\frac{dy}{dx} = 6x^2, \quad y(0) = 1.$

Step 1. The right side already holds no y, so separation is just $dy = 6x^2\,dx$.
Step 2. Integrate both sides. The antiderivative of the right side is $\boxed{2x^3}$, so $y = 2x^3 + C$.
Step 3. Apply $y(0) = 1$: $1 = 2(0)^3 + C$, so $C = 1$ and the particular solution is $y = 2x^3 + 1$.
Step 4. $y(2) = 2(8) + 1 = \boxed{17}$
Note: the $+C$ goes on one side only. Writing C_1 and C_2 on both sides is not wrong, but it is one more thing to carry.

2. $\frac{dy}{dx} = 3x^2 - 4, \quad y(2) = 5.$

Step 1. Separate: $dy = (3x^2 - 4)\,dx$.
Step 2. Integrating the right side gives $\boxed{x^3 - 4x}$, so $y = x^3 - 4x + C$.
Step 3. Apply the condition at $x = 2$: $5 = 8 - 8 + C$, which gives $\boxed{C = 5}$
Step 4. Particular solution: $y = x^3 - 4x + 5$.
Listen for: students who leave the answer as $y = x^3 - 4x + C$. An initial-value problem is not finished until the constant is a number.

3. $\dfrac{dy}{dx} = xy, \quad y(0) = 2.$

Step 1. Both variables appear, so separate properly: $\dfrac{dy}{y} = x\,dx$.

Step 2. Integrate. The right side gives $\boxed{x^2/2}$, and the left gives $\ln|y|$, so $\ln|y| = \dfrac{x^2}{2} + C$.

Step 3. Exponentiate: $y = Ae^{x^2/2}$ where $A = e^C$. Applying $y(0) = 2$ gives $2 = Ae^0$, so $A = 2$ and $y = 2e^{x^2/2}$.

Step 4. $y(2) = 2e^2 = \boxed{14.78}$ (to two decimal places).

Common wrong answer: 7.39 — the initial condition was never applied, leaving the coefficient at 1.

4. $\dfrac{dy}{dx} = \dfrac{2x+1}{y}, \quad y(0) = 3.$

Step 1. Cross-multiply into separated form: $y\,dy = (2x+1)\,dx$.

Step 2. Integrate: $\dfrac{y^2}{2} = x^2 + x + C$.

Step 3. Apply $y(0) = 3$ *now*, before solving for y: $\dfrac{9}{2} = 0 + 0 + C$, so $\boxed{C = \frac{9}{2}}$

Step 4. Multiply through by 2: $y^2 = 2x^2 + 2x + 9$, and the initial value is positive, so $y = \sqrt{2x^2 + 2x + 9}$. Then $y(2) = \sqrt{21} = \boxed{4.58}$

Common wrong answer: 3.24 — the equation was not multiplied by 2 before taking the square root.

5. $\dfrac{dy}{dx} = y^2, \quad y(1) = 1.$

Step 1. Separate: $\dfrac{dy}{y^2} = dx$, that is $y^{-2}\,dy = dx$.

Step 2. The left side integrates to $\boxed{-1/y}$, so $-\dfrac{1}{y} = x + C$.

Step 3. Apply $y(1) = 1$: $\quad -1 = 1 + C$, so $C = -2$ and $-\dfrac{1}{y} = x - 2$, giving $y = \dfrac{1}{2-x}$.

Step 4. $y(1.5) = \dfrac{1}{0.5} = \boxed{2}$

Step 5 (domain). The solution blows up at $x = 2$. The initial point $x = 1$ lies to the left of that, so the particular solution is defined on $x < 2$ — a solution of an IVP lives on the interval containing the initial point, not on every x where the formula happens to evaluate.

6. Radioactive decay: 80 **mg, half-life** 10 **days.**

Step 1. $\dfrac{dA}{A} = -k\,dt$ integrates to $\ln A = -kt + C$, so $A = A_0 e^{-kt}$ with $A_0 = 80$ mg.

Step 2. Half-life: $40 = 80e^{-10k}$, so $e^{-10k} = \frac{1}{2}$ and $k = \dfrac{\ln 2}{10} \approx 0.069315$ per day.

Step 3. After 25 days: $A(25) = 80e^{-(\ln 2/10)(25)} = 80 \cdot 2^{-2.5}$.

Step 4. $A(25) = \boxed{14.14 \text{ mg}}$

Common wrong answer: 20 mg — 25 days was rounded down to two whole half-lives. Two and a half half-lives is not two.

7. $\dfrac{dy}{dx} = y\cos x, \quad y(0) = 4.$

Step 1. Separate: $\dfrac{dy}{y} = \cos x\, dx$.

Step 2. The right side integrates to $\boxed{\sin(x)}$, so $\ln|y| = \sin x + C$.

Step 3. Exponentiate to $y = Ae^{\sin x}$, and $y(0) = 4$ with $\sin 0 = 0$ gives $A = 4$. Particular solution: $y = 4e^{\sin x}$.

Step 4. At $x = 1$ radian, $\sin 1 \approx 0.841471$, so $y(1) = 4e^{0.841471} = \boxed{9.28}$

Common wrong answer: 1.72 — cosine was integrated to $-\sin x$. Differentiation introduces the minus sign; integration does not.

8. $\dfrac{dy}{dx} = \dfrac{3x^2}{2y}, \quad y(1) = 2.$

Step 1. Separate: $2y\, dy = 3x^2\, dx$.

Step 2. The left side integrates to $\boxed{y^2}$ and the right to x^3, so $y^2 = x^3 + C$.

Step 3. Apply $y(1) = 2$: $\quad 4 = 1 + C$, so $C = 3$ and $y^2 = x^3 + 3$. The initial value $y = 2$ is positive, which selects the positive branch: $y = \sqrt{x^3 + 3}$.

Step 4. $y(3) = \sqrt{27 + 3} = \sqrt{30} = \boxed{5.48}$

Why the branch matters: $y = -\sqrt{x^3 + 3}$ satisfies the same differential equation, but not the initial condition. The condition picks one of the two branches, and the solution never crosses $y = 0$.

9. Population: $\dfrac{dP}{dt} = 0.08P, \quad P(0) = 500.$

Step 1. $\dfrac{dP}{P} = 0.08\, dt$ integrates to $\ln P = 0.08t + C$, so $P = 500e^{0.08t}$ after applying $P(0) = 500$.

Step 2. Set $P = 1500$: $\quad 1500 = 500e^{0.08t}$, so $e^{0.08t} = 3$. Note that only the *ratio* 1500/500 survives.

Step 3. Take logarithms: $0.08t = \ln 3$, so $t = \dfrac{\ln 3}{0.08}$.

Step 4. $t = \boxed{13.73 \text{ years}}$

Common wrong answer: 91.42 years — $\ln 1500$ was used instead of $\ln 3$, because the initial population was not divided out first.

10. Logistic equation $\frac{dy}{dx} = y(3-y)$**.** *Part (b) is an open construction and explanation, flagged for manual review; the model answer is below.*

Step 1 (part a). Equilibrium solutions are the constant solutions, where $\frac{dy}{dx} = 0$ for all x. Setting $y(3-y) = 0$ gives $\boxed{y = 0 \text{ and } y = 3}$

Step 2 (the slope field). The slope depends only on y, so every segment in a horizontal row is parallel — that is the fastest way to fill the lattice. Row by row: at $y = -1$ the slope is $(-1)(4) = -4$ (steeply falling); at $y = 0$ and $y = 3$ it is 0 (horizontal); at $y = 1$ it is $(1)(2) = 2$ (rising); at $y = 2$ it is $(2)(1) = 2$ (rising); at $y = 4$ it is $(4)(-1) = -4$ (steeply falling).

Step 3 (long-run behaviour). The solution through $(0,1)$ starts between the two equilibria, where the slope is positive, so it increases. It can never cross $y = 3$, because that would require crossing an equilibrium solution. Therefore it rises and approaches $y = 3$ asymptotically as x grows — $y = 3$ is the carrying capacity.

Full credit: both equilibria found in (a); segments drawn horizontal along $y = 0$ and $y = 3$, up-sloping strictly between them and down-sloping outside; and a sentence saying the solution increases toward 3 without reaching it. A sketch whose segments cross $y = 3$ is the error to look for.